

Measuring Collision Anxiety in XR Exergames

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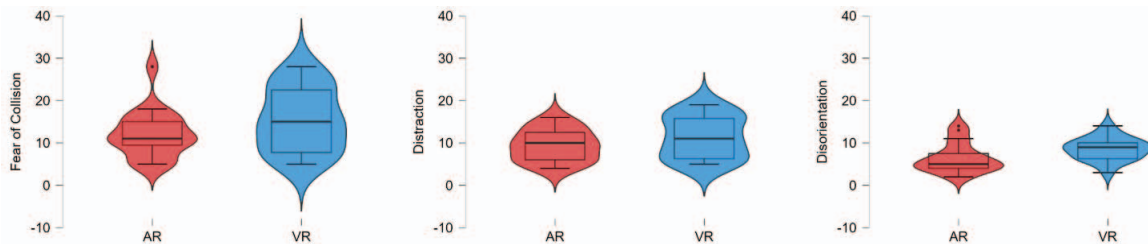
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Figure 1: Violin plots showing differences in *Collision Anxiety* between an AR and VR version of a game measured with our newly developed questionnaire. Differences become visible in all three sub-scales, *fear of collision*, *distraction* and *disorientation*.

ABSTRACT

Extended Reality (XR) applications have become increasingly relevant due to technical advances, but also have their own challenges regarding the user's real-world orientation. We propose a novel definition and a not yet validated questionnaire for the feeling of disorientation and fear of colliding with real objects in XR, called *Collision Anxiety* (CA). Participants ($N = 37$) played an AR and VR version of a game with results suggesting that while an AR game can provide a similar player experience compared to its VR equivalent, differences regarding CA exist.

Index Terms: Human-centered computing; Applied computing—Interaction design—Empirical studies in interaction design

1 COLLISION ANXIETY IN XR GAMING

The new generation of wireless HMDs offer a great range of movement but also carry the risk of colliding with the environment, especially when using AR and VR games at home. While most current VR devices completely cover the players' field of view and immerse the user in a virtual world, AR enables the users to see their surroundings. Especially in VR, users might worry about colliding with the environment, especially when using roomscale applications in confined spaces.

We propose the term collision anxiety (CA) for the *fear of colliding with or hitting an object while using an XR application*. Currently, empirical research on CA is very limited: To our knowledge, there is only one study with a focus on anxiety relief through exergaming where CA is briefly discussed and measured by a single question [5]. Another paper has reported the participants' "fear of colliding with physical objects in real life" [1], which was grouped under *psychophysical discomfort*, along with motion sickness. Here, fear of collision was deduced from participants' answers in semistructured interviews. A work on obstacle avoidance used a self-developed questionnaire with five items asking about fear while walking and sense of distance to obstacles [2]. Finally, a four-item scale on fear as an emotional state has been used to assess discomfort

in VR [3]. According, it has been remarked that fear of collision needs further research [1].

Being able to directly measure CA will become increasingly more relevant in the future the more XR devices enable the user to move freely. Even if no collisions with the environment are occurring, the users might still experience CA. Furthermore, in order to make good design decisions, especially in XR gaming, there is the need to distinguish between the impact of other variables (such as immersion) and CA. Hence, we developed a questionnaire to capture CA and tested it on a self-developed exergame with comparable AR and VR versions. In order to delimit CA from other variables, presence, immersion, and motivation were reported as well.

2 MEASURING COLLISION ANXIETY

2.1 Collision Anxiety Questionnaire

To measure the experienced CA a questionnaire (CAQ) with twelve items was developed. The items are divided into four sub-scales. The first sub-scale *physical collision* consists of a single item asking whether a collision has occurred during playing. The sub-scale *fear of collision* includes all items regarding the fear of hitting an object or getting injured due to being unaware of one's own position within the real room. The third sub-scale *distraction* addresses to what extent the immersion is disrupted due to players focusing on avoiding collisions. The last sub-scale *disorientation* asks to what extent the players were aware of their own and surrounding objects' positions during playing. All items are rated on a scale ranging from "strongly disagree" (1) to "strongly agree" (7) with some items being reversed. A higher score (overall and on each sub-scale) indicates higher collision anxiety.

2.2 ExARcism in AR and VR

To test the CAQ, the exergame *ExARcism* was developed using the game engine Unity and a Meta Quest 2. Exergames are a sub-genre of video games in which the player's body functions as an input device. They offer an increasingly relevant approach towards meeting people's need for at-home fitness activities by combining physical activity with a highly immersive gaming experience, increasing enjoyment and motivation [4].

Players play in first-person perspective inside a mansion haunted by ghosts. The game has two alternating phases that are completed two times: In the *Drumming Phase* (75 seconds), players have to use the Meta Quest 2 controller to hit one of four drums, that are hovering in different locations inside the room. To force players to

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	Sub-scales	Items	Scale
CA1	Physical Collision	I collided with an object from the real environment during the study.	Y/N
CA2	Fear of Collision	I was afraid of hitting an object in the real room.	7-point likert scale
		I was afraid of running into an object in real room.	7-point likert scale
		I was not afraid of getting hurt while playing.	7-point likert scale
		I felt uncomfortable because I did not know where objects were in the real room.	7-point likert scale
		I felt uncomfortable because I did not know where I was in the real room.	7-point likert scale
CA3	Distraction	I was focused on not colliding with objects in the real room.	7-point likert scale
		I could not engage with the virtual world because I had to pay attention to the real room.	7-point likert scale
		I moved through space during the game, as I would in a real room.	7-point likert scale
		I held back while playing to avoid a collision/(to avoid hurting myself).	7-point likert scale
CA4	Disorientation	I knew at any time where I was in the real room.	7-point likert scale
		I knew at any time how far away I was from objects in the real room.	7-point likert scale

Table 1: The proposed Collision Anxiety Questionnaire (alpha CAQ).

move within the play area, they all activate and deactivate one after another at fixed times. In the *Ghost Hunting Phase* (90 seconds), the players have to capture the ghosts with the controllers and exorcise them by throwing them through a portal.

The players could only see the virtual environment in the VR version. In the AR version, players could see their surroundings through the *passthrough mode* of the Meta Quest 2 with relevant objects like the drums, portal, and ghosts, as well as decorations projected into the monochrome camera feed. A grayscale-to-color mapping was used to make the game experience more aesthetically pleasing and create a similar environment compared to the VR version as seen in Figure 2.

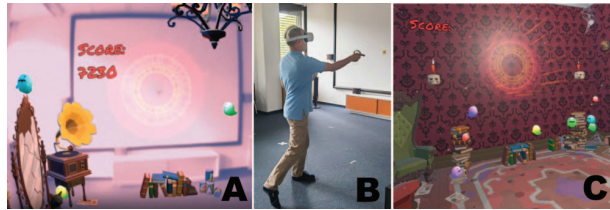


Figure 2: **A** The AR game environment, **B** the lab showing a participant and **C** the VR game environment showing the portal during the *Ghost Hunting Phase*.

2.3 Lab Study

A total of 37 participants completed the study, mostly university students as reflected in the age demographics for this sample ($M = 24.87$, $SD = 3.00$) with more women ($n = 23$) than men ($n = 14$). Most participants had prior experience with exergames ($n = 28$) and only two reported having no prior experience with VR technology. Participants were randomly assigned to one of the experimental conditions and played either the AR or the VR version of the *ExArcism* game. Afterward, players were asked to fill out the CAQ as well as questionnaires regarding player experience for delimitation from CA.

Regarding the difference in CA between the VR and AR condition, a significant difference was found in the *disorientation* sub-scale of the CAQ. The 18 participants who played the VR version of the exergame ($M = 4.28$, $SD = 1.32$) demonstrated significantly higher scores for this sub-scale ($t(35) = -2.16$, $p = .019$), compared to the 19 participants who played the AR version ($M = 3.18$, $SD = 1.72$). No significant difference was demonstrated regarding the *distraction*

from the virtual environment or the *fear of collision*.

The violin plots in Figure 1 show the noticeable descriptive differences for all three sub-scales between the AR and the VR condition. No significant differences have been found for player experience between both versions.

3 FUTURE RESEARCH & CONCLUSION

The CAQ worked well in the first trial as the mean differences in all three sub-scales indicate a possible difference in CA between players of the AR and VR version with a significant difference demonstrated for the disorientation sub-scale. Although these findings need to be interpreted with caution since we only had a limited number of participants, the CAQ seems a promising tool for measuring CA in XR applications. Meanwhile, player experience did not differ significantly suggesting that differences were due to CA. The CAQ is work in progress and has not yet been validated. Further research is being conducted regarding smaller spaces, collisions with real or virtual objects and the influence of calibration on CA. Another question is whether collision anxiety will change over time as users become more comfortable in the VR environment [1]. Following, we plan to potentially reduce the items, conduct a factor analysis to validate the internal consistency and also aim for an analysis of possible interdependencies with related constructs like presence and immersion to enable a clear delimitation. This work in progress contributes to the future design of XR applications as the construct of CA could be of substantial help to explore factors to ensure a good and safe user experience.

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